



Effect of Inflation and Exchange Rates on Financial Performance of Selected Quoted Manufacturing Companies in Nigeria

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Abstract

The financial performance of manufacturing companies in Nigeria has been increasingly unstable, marked by fluctuating profitability, declining returns, and rising costs. This instability is largely attributed to persistent inflation and exchange rate volatility, both of which have intensified in recent years. The study examines the effect of inflation and exchange rates, on the financial performance of selected quoted manufacturing companies in Nigeria, using Tobin's Q as a proxy for firm performance. The study adopts ex-post factor research design and covered 43 quoted manufacturing firms in Nigeria. Utilizing panel data from 2014 to 2023 which was extracted from published annual reports of Statistical Bulletin of the Central Bank of Nigeria, National Bureau of Statistics and Nigerian Stock Exchange and applied regression analysis through the least squares method. The study found that both inflation and exchange rates have negative and significant effect on financial performance (Tobin's Q) of selected quoted manufacturing firms in Nigeria 5% significant level. The study concludes that inflation and exchange rates negatively influence financial performance (Tobin's Q) of selected quoted manufacturing firms in Nigeria. the study therefore, recommend that manufacturing companies in Nigeria should strengthen their resilience to rising

inflation-related production costs by enhancing operational efficiency through energy-efficient technologies and waste reduction. The study also recommends that manufacturing companies in Nigeria should implement strong foreign exchange risk management strategies, such as hedging with forward contracts or options, to protect against exchange rate volatility.

Keywords: Exchange Rate, Financial Performance, Inflation Rate, Manufacturing Companies, Tobin's Q

Introduction

The financial performance of manufacturing companies is critical to economic stability and growth, especially within emerging markets. Globally, the manufacturing sector contributes significantly to GDP and employment in both developed and developing nations. In the United States, Europe, and parts of Asia, manufacturing companies play a central role in national economies, continuously adapting to market demands and technological advancements to remain competitive (Smith & Brown, 2022). Financial performance, often measured by metrics like return on assets (ROA), return on equity (ROE), and profit margins, reflects a company's ability to manage its resources and generate returns for stakeholders. In Africa, the manufacturing sector has shown potential for economic transformation. However, many African countries face financial performance challenges due to factors such as political instability, limited access to finance, and fluctuating commodity prices (Akpan & Johnson, 2021). These challenges are

further compounded by external economic shocks, making it difficult for African manufacturing companies to maintain stable financial performance.

In Nigeria, manufacturing is an essential sector, accounting for a significant portion of GDP and employment. Quoted manufacturing companies, which are listed on the Nigerian Stock Exchange, contribute to capital market development by attracting local and foreign investment. However, the sector faces numerous challenges, including high production costs, weak infrastructure, and market competition. The financial performance of these companies has been unstable, partly due to external factors like inflation and fluctuating exchange rates. For instance, the high inflation rate in Nigeria, driven by factors like rising energy costs and food prices, affects the purchasing power of consumers and the cost structure of manufacturing firms (Okafor & Eze, 2022). When inflation is high, the cost of raw materials, wages, and operating expenses increases, squeezing profit margins and limiting

growth potential for manufacturing firms.

Additionally, the volatility of the Nigerian Naira's exchange rate against major currencies such as the U.S. Dollar has significant implications for the financial performance of manufacturing companies, particularly those reliant on imported raw materials (Adebayo & Ojo, 2023). Fluctuations in exchange rates lead to higher costs of imported goods and services, making it challenging for firms to manage costs effectively. A depreciating Naira increases the cost of imports, affecting cash flows and profitability for companies in the manufacturing sector. This volatility also discourages foreign investment, as investors seek stable environments for returns. Exchange rate instability, when combined with high inflation, creates an uncertain economic environment that can undermine the profitability and sustainability of these firms (Ogunleye & Akinola, 2022).

Understanding the effects of inflation and exchange rate volatility on the financial performance of quoted manufacturing companies in Nigeria is essential for policy formulation and corporate decision-making. By analyzing how these economic factors impact profitability, liquidity, and overall performance, stakeholders can develop strategies to mitigate risks and enhance the sector's resilience.

The financial performance of manufacturing companies in Nigeria has been increasingly unstable, marked by fluctuating profitability, declining returns, and rising costs. This instability is largely attributed to persistent inflation and exchange rate volatility, both of which have intensified in recent years. Inflation, characterized by the continuous rise in prices, affects the cost structure of manufacturing companies by increasing expenses on raw materials, labor, and utilities. This impact, combined with rising inflationary pressures in Nigeria, constrains the profitability of manufacturing firms, which struggle to adjust their pricing without losing competitiveness in the market (Okafor & Eze, 2022).

Moreover, the volatility of the Nigerian Naira against major global currencies, especially the U.S. Dollar, further complicates the financial landscape for manufacturing firms. Companies reliant on imported materials face unpredictable cost increases as the Naira weakens, affecting their ability to manage cash flows and profit margins (Adebayo & Ojo, 2023). Exchange rate fluctuations also raise the risk of foreign currency liabilities, increasing financial costs and discouraging long-term investment in the sector. Despite various government interventions aimed at stabilizing the currency, exchange rate instability continues to hamper the manufacturing sector's financial performance and restricts access to affordable foreign capital (Ogunleye & Akinola, 2022).

Although previous studies have examined the general impacts of macroeconomic variables on financial performance, there is limited research specifically focused on how inflation and exchange rate fluctuations jointly affect the financial stability of

quoted manufacturing companies in Nigeria. Most prior studies in this area have primarily examined isolated variables or limited scopes, often within foreign contexts that may not be directly applicable to Nigeria's unique economic landscape. For instance, Muhammad (2015) investigated fiscal policy and its impact on economic growth in Nigeria, while Baba and Nasieku (2016) analyzed the relationships among real interest rates, unemployment, and exchange rates within Nigerian commercial banks. In other contexts, Williams (2018) assessed the effects of exchange rate fluctuations on corporate performance, and Musah et al. (2018) studied the influence of interest rate spreads on bank profitability in Ghana. Commonly, these studies have employed financial metrics such as Return on Assets (ROA), Return on Equity (ROE), or Return on Investment (ROI) to evaluate financial performance (Nnajeze et al., 2021; Musah et al., 2018; Madana, 2022).

It is in the light of the above that this study aims to fill this gap by examining the combined effect of inflation and exchange rates on the financial performance of selected quoted manufacturing companies in Nigeria focusing on Tobin Q as a proxy financial performance.

H₀₁: Inflation rate has no significant effect on financial performance (Tobin's Q) of selected quoted manufacturing companies in Nigeria

H₀₂: Exchange rate has no significant effect on financial performance (Tobin's Q) of selected quoted manufacturing companies in Nigeria

Literature Review

Financial Performance

Martínez-Caro et al. (2021) describe financial performance as an evaluation of a company's financial health, demonstrated through its capacity to create shareholder value via effective asset management, profitability, and liquidity. It signifies the company's success in converting resources into financial results that promote sustainable growth. Similarly, Agyemang and Ansong (2021) define financial performance as the overall outcome achieved by a firm in terms of profitability, liquidity, and solvency, focusing on maximizing stakeholder wealth. This concept encompasses short-term operational efficiency and long-term sustainability, both essential for corporate success.

Yunus and Santoso (2022) view financial performance as the effectiveness of a company's strategies in generating revenue and maintaining profitability, reflecting the company's ability to maximize shareholder value through efficient revenue management and cost control. High financial performance suggests sustainable operations in competitive markets. It is often measured using market-based indicators such as stock prices, return on investment (ROI), and Tobin's Q, as it mirrors both the firm's internal financial health and external market confidence (Nguyen et al., 2023).

Iswatia and Anshoria (2017) regard financial performance as a measure of an organization's ability to acquire and manage resources to build a competitive advantage. Similarly, Kithinji (2020) emphasizes that financial performance reflects a firm's efficiency in utilizing its resources to generate returns for investors. Common metrics for assessing financial performance include Return on Equity (ROE), Return on Assets (ROA), and Tobin's Q, which evaluate profitability, growth, and market confidence. Given this framework, this study measured financial performance using Tobin's Q.

Tobin's Q

Singh and Sharma (2022) define Tobin's Q as the ratio of a company's market value to the replacement cost of its assets, offering a perspective on how well the market values a company's assets in relation to their actual cost. A Tobin's Q ratio greater than 1 indicates that the market values the firm's assets higher than their replacement cost, signaling growth potential and investor confidence. This ratio serves as an indicator of market efficiency, with values around 1 suggesting an efficient valuation of a firm's assets. Ratios significantly above or below 1 may point to market inefficiencies, where the firm could be overvalued or undervalued relative to its asset replacement costs (Le & Vo, 2022).

Chen and Su (2023) describe Tobin's Q as a financial metric that reflects a firm's investment opportunities by comparing its market value to the replacement cost of physical assets. A high Tobin's Q indicates investor confidence in the company's future profitability and value creation. Tobin's Q is widely applied as a firm performance indicator, encompassing both financial and operational efficiency. By combining the internal measure of asset replacement cost with external market valuation, Tobin's Q offers a comprehensive view of a firm's capacity to create shareholder value (Rahman & Ali, 2023).

Tobin's Q is regarded as a valuable tool for assessing financial performance, as it integrates both a firm's internal operational efficiency and the market's external valuation. A Q ratio above 1 suggests that the market perceives the firm's assets as valuable, reflecting growth potential and investor confidence, while a ratio below 1 may indicate undervaluation or operational inefficiency (Singh & Sharma, 2022; Chen & Su, 2023). As a forward-looking measure, Tobin's Q is beneficial for evaluating long-term investment potential and guiding strategic decision-making within firms (Le & Vo, 2022; Rahman & Ali, 2023).

Inflation Rate

According to Nguyen (2015), inflation rate is defined as the rate of change in the general price level of goods and services in an economy, typically measured by the

consumer price index (CPI). Inflation rate is the percentage change in the price level of a basket of consumer goods and services, which is commonly measured by the consumer price index (CPI) (Inam & Udoh, 2019). According to Bittencourt (2012), Inflation rate is the rate at which the general price level of goods and services in an economy increases over a given period of time, typically measured by the year-over-year change in the consumer price index (CPI). Inflation rate is the rate of change in the general price level of a basket of goods and services consumed by households, as measured by the consumer price index (CPI) (Asongu, 2013). Inflation rate is the sustained increase in the general price level of goods and services in an economy over time, which leads to a decline in the purchasing power of the currency (Epaphra & Massawe, 2017).

Inflation is the rate at which the general price level of goods and services in an economy increases over time (Mishkin, 2016). Inflation refers to the sustained increase in general price level of goods and services in an economy over time (Mankiw, 2019). It is typically measured by indices such as the Consumer Price Index (CPI) or the Producer Price Index (PPI). High inflation can erode the purchasing power of consumers, leading to a decline in real wages and standards of living. Conversely, low or stable inflation is generally associated with economic stability and growth. The inflation rate is measure of the rate at which the general price level of goods and services in an economy increases over time (Mankiw, 2021). It is a crucial macroeconomic factor that can have significant impacts on consumer purchasing power, investment decisions, and the overall economic stability of a country.

One economic variable that has continued to truncate or cause setback to many countries is inflation because during inflation, the value of money declines as the prices of goods and services increase (Maimunah & Patmati, 2018). Since inflation affect profit and others performance variables like net asset, return on equity etc., the first point of reaction is from sales volume and also influencing the level of operating costs which also result to decline in the net asset per share and other variables. The resultant effect is that, the rise in inflation causes a decline in the value of currencies and this will affect companies that utilize local components or materials for production because more naira are used to purchase few components or materials. When the inflation rate is low, companies can still operate efficiently within a certain level, but high inflation impedes companies' performance (Akabom-Ita, 2017).

Exchange Rate

The exchange rate is the price of one currency expressed in terms of another currency (Mankiw & Taylor, 2021). This definition highlights that the exchange rate is the relative price between two different currencies, indicating how much of one currency is needed to purchase a unit of another currency. The exchange rate is the rate at which

one currency is exchanged for another (Krugman et al., 2018). This definition emphasizes the notion of an exchange, where one currency is traded for another at a specific rate. The exchange rate is the price of one country's currency in terms of another country's currency (Husted & Melvin, 2022). This definition focuses on the exchange rate as the price of a domestic currency relative to a foreign currency.

The exchange rate is the price at which one currency can be converted into another (Obstfeld & Rogoff, 2021). This definition highlights the concept of currency conversion, where one currency is exchanged for another at a specific rate. The exchange rate is the number of units of a country's currency that can be exchanged for one unit of another country's currency (Copeland, 2020). This definition emphasizes the quantitative aspect of the exchange rate, indicating the number of units of one currency required to purchase a single unit of another currency. The exchange rate is the value of one currency in terms of another currency (Madura, 2021). This definition underscores the notion of the value of a currency relative to another currency, which is reflected in the exchange rate. The exchange rate is the rate at which one currency is traded for another (Dornbusch et al., 2018). This definition focuses on the trading aspect of exchange rates, where currencies are exchanged at a specific rate. The exchange rate is the price of a currency in terms of another currency" (Salvatore, 2023). This definition emphasizes the exchange rate as the price of one currency expressed in terms of another currency.

Exchange rates are the prices at which one currency can be changed for another (Krugman et al., 2018). They represent the value of country's currency relative to another country's currency and are determined by the supply and demand for different currencies in the foreign exchange market (Mankiw & Taylor, 2021). Exchange rates play a crucial role in international trade and finance, as they affect the relative prices of goods and services between countries, the flow of capital and investment, and purchasing power of individuals and businesses (Husted & Melvin, 2022).

Exchange rates can be influenced by various economic factors, such as inflation, interest rates, economic growth, and political stability (Obstfeld & Rogoff, 2021). Governments and central banks may also intervene in the foreign exchange market to manage the value of their currency and achieve economic policy objectives (Copeland, 2020). Understanding and predicting exchange rate movements is crucial for individuals, businesses, and policymakers engaged in international transactions (Madura, 2021)

Empirical Review

Inflation Rate and Financial Performance

Imegi, et al. (2023) examined the impact of inflation on the capital structure of industrial goods manufacturing firms in Nigeria. The study focused on how inflation,

nominal, and real interest rates influence the debt-to-equity ratio, using panel data from 2012-2021. Data was collected from the Central Bank of Nigeria and the firms' financial statements, and analyzed with panel regression models. Findings from the fixed effects model indicated that while real interest rate changes could explain a 45% variation in the debt-to-equity ratio, inflation did not have a statistically significant impact. Criticism of the study lies in its narrow scope, focusing only on Nigeria and limiting broader applicability. The study suggests that firms with high debt levels should reduce debt to manage costs effectively.

Scemeleva & Nagornova (2023) investigated the relationship between inflation and firm performance across Central and Eastern European (CEE) countries. This bachelor's thesis utilized firm-level data from 2005 to 2021, sourced from the ORBIS database, to assess profitability and liquidity responses to inflation. Regression analysis showed a positive impact of inflation on profitability, particularly for anticipated inflation, whereas unanticipated inflation had a smaller effect. A noted limitation is that the study's focus on the CEE region restricts generalizability to other economic contexts. The findings highlight inflation's differentiated effects based on its demand-pull versus cost-push nature, offering policy implications for inflation management.

Tërstena et al. (2023) analyzed inflation's impact on business profitability in Kosovo, focusing on factors like profit tax and GDP. Using a quantitative design, secondary data from 120 businesses' financial statements (2018-2021) were collected. Descriptive and linear trend analyses revealed a positive relationship between inflation and profitability, suggesting that businesses benefit from inflation-related pricing adjustments. A critique of this study is its limited scope, relying solely on Kosovo-based businesses, which limits broader application. The findings support the notion that inflation can benefit firms by enhancing profit margins amid rising costs.

Bilalli & Sadiku (2023) explored the effect of inflation on financial sector performance in Western Balkan countries, using World Bank data from 2002 to 2021. The study applied various panel regression methods, including OLS and GMM, to assess inflation's influence on credit to the private sector and money supply as GDP ratios. Results showed that inflation correlates positively with private sector credit but negatively with broad money. Limitations include potential regional bias and over-reliance on macro-level data. This study underscores inflation's differential impacts within financial sectors, influencing both money supply and private sector credit access

Moyo and Tursoy (2020) examined the impact of inflation and exchange rate on the financial performance of commercial banks in South Africa. The study employed a quantitative research approach and utilized secondary data from the four largest commercial banks in South Africa. The population for the study was the commercial

banks covering the period from 20003 to 2019. The study collected secondary data from the annual reports and financial statements of the selected commercial banks, as well as data from the South Africa Reserve Bank and other relevant sources. The study used the Autoregressive Distributed Lag (ARDL) model, the Fully Modified Ordinary Least Squares (FMOLS) model, and the Dynamic Ordinary Least Square (DOLS) model to analyze the data and examine the relationships between the variables. The findings of the study illustrated that there is a significant inverse relationship between inflation and the return on equity and there is a weak relationship between exchange rate and the return on equity. The study focused only on the four largest commercial banks in South Africa, which may not fully represent the entire commercial banking industry in the country, therefore, the finding from the study cannot be generalized to other banks in South Africa and to other Countries of the World. Also, the study did not consider the potential impact of other macroeconomic factors, such as GDP growth, interest rates, or regulatory changes, which could also influence the financial performance of commercial banks.

Baba & Ashogbon (2019) examined inflation's effect on the financial performance of Nigerian commercial banks over 2006-2015 using panel data regression. The study focused on return on equity (ROE) as the performance metric, finding that inflation has an insignificant effect on ROE in the banking sector. Data for 23 banks was analyzed, revealing minimal variation in ROE over the study period. The study's limitation lies in its narrow focus on Nigeria, which limits applicability to other regions. Findings suggest that inflation may not directly impact bank profitability, challenging assumptions about inflation's universal effects on financial performance. Alimi (2014) investigated inflation's long-term impact on Nigeria's financial sector, covering data from 1970 to 2012. The study utilized broad money, quasi money, and private sector credit as proxies for financial sector development, analyzing their relationships with inflation. Using econometric analysis, the study found that inflation negatively affects financial sector performance, suggesting that high inflation rates reduce resource allocation efficiency. A critique is the study's historical data reliance, which may not reflect current economic dynamics. This research highlights the need for stable inflation to foster a productive financial sector conducive to economic growth

Exchange Rate Financial Performance

Mate et al. (2022) conducted a study on the effect of foreign exchange rates on the financial performance of listed manufacturing companies in Kenya. This study aimed to assess the influence of exchange rates, alongside interest rates and inflation, on the profitability of the manufacturing sector, which has struggled to meet growth targets. Using a descriptive correlational research design and panel data from 2015 to 2020,

the researchers analyzed data from eight listed firms. Their findings indicated that while economic growth and inflation positively impacted performance, the exchange rate had a negative effect, with a 1% increase in exchange rate reducing performance by 1.532%. The study highlighted the importance of stable macroeconomic policies to support manufacturing performance. However, a limitation noted was the short time frame, which may not capture long-term effects of economic fluctuations on the manufacturing sector.

Orji & Ezeanyaeji (2022) explored the effect of exchange rates on Nigeria's manufacturing sector from 1990 to 2020, employing a canonical cointegrating regression (CCR) model. They found that exchange rate devaluation and fluctuations adversely impact manufacturing output. The study recommends a stable exchange rate policy and cautions against frequent policy changes, which destabilize the sector. One critique is the broad timeframe, which may encompass outdated data. Findings underscore the necessity of aligning exchange rate policies with the manufacturing sector's needs for sustainable growth.

Osho & Efuntade (2019) examined the impact of exchange rate fluctuations on the financial performance of multinational companies in Nigeria. Their study, grounded in theories such as transaction cost and inflation theories, utilized secondary data from the Central Bank of Nigeria and company reports. Applying the Ordinary Least Square Linear Regression model, they found a significant adverse effect of exchange rate fluctuations on corporate performance. The study suggests that multinational companies should adopt comprehensive foreign exchange risk management strategies. A critique of this study is the reliance on secondary data, which may not capture real-time impacts of exchange fluctuations on operational costs.

Olabode (2018) analyzed the effect of foreign exchange rates on the performance of Nigeria's manufacturing sector from 1990 to 2016. Using time series data and employing OLS and co-integration techniques, the study found significant negative impacts of exchange rates, inflation, and interest rates on manufacturing performance. The study's limitation lies in its focus on historical data, potentially lacking relevance in current contexts. Recommendations included exchange rate stabilization policies to enhance manufacturing sector stability and growth.

Agubata & Odubuasi (2018) focused on how exchange rate fluctuations influence the financial performance of food, beverage, and tobacco manufacturing firms in Nigeria from 2005 to 2014. Utilizing an ex post facto research design and multiple regression, the study showed positive but insignificant effects of exchange rates on return on assets and equity. A criticism is the study's restriction to a few manufacturing sectors, limiting its applicability to other sectors. Findings suggest a government-driven provision of low-interest credit facilities to aid firms' resilience against exchange rate volatility.

Majumdar (2017) explored the effects of macroeconomic variables, particularly exchange rates, on the financial performance of Nigerian firms. Using a longitudinal data approach, the study analyzed exchange rate policies from the fixed rates of post-independence Nigeria to the floating regime post-1986. The findings showed that fluctuating exchange rates adversely affected firms by increasing operational costs, especially for companies reliant on imports. The study underscored the need for stable exchange rate policies to support business continuity. A primary limitation noted was the broad focus on multiple macroeconomic variables, which may dilute the specific impact of exchange rates on firm-level performance.

Danish (2012) analyzed how exchange rate instability influences financial intermediation and the balance of payments in developing countries. Through a cross-sectional study of multiple economies, including Nigeria, the research demonstrated that exchange rate volatility increases the cost of imported goods, thereby reducing firm profitability in import-heavy industries. Using econometric modeling, the finding of the study showed that currency volatility discourages long-term investments. The study, however, faced criticism for its limited focus on developing countries, thus limiting comparisons with more developed economies

Theoretical Framework

Keynesian Theory

This study is grounded in Keynesian Economic Theory, a framework pioneered by John Maynard Keynes in 1936 through his seminal work, *The General Theory of Employment, Interest, and Money*. Keynes developed his theory during the Great Depression, a period when classical economic models could not account for severe unemployment and economic stagnation. Keynes posited that aggregate demand, or total spending within an economy, is the main driver of economic output and employment levels. According to Keynes, insufficient aggregate demand could result in prolonged recessions, thereby justifying government intervention through fiscal measures such as increased public spending and tax modifications to stimulate demand, spur economic growth, and reduce unemployment (Keynes, 1936).

Keynesian theory was further refined by economists like Paul Samuelson and John Hicks. Samuelson's interpretation, combined with Hicks' development of the IS-LM model, provided a formalized approach to understanding the interaction between interest rates, national income, and monetary policy (Samuelson, 1948; Hicks, 1937). Later advancements by New Keynesian economists, including Gregory Mankiw and David Romer, incorporated concepts like price stickiness and imperfect competition, addressing limitations in the original framework and enhancing its relevance to modern economic issues (Mankiw, 1990; Romer, 1993). However, Keynesian theory has also faced notable criticism, particularly from monetarists such as Milton

Friedman, who argued that excessive government intervention can lead to inflation and economic inefficiencies. Instead, Friedman emphasized the importance of controlling the money supply to manage economic cycles more effectively (Friedman, 1968).

Despite the criticisms, the theory is highly relevant to the study of inflation and exchange rates on the financial performance of selected quoted manufacturing companies in Nigeria, as it provides a framework for understanding how aggregate demand, driven by factors like consumer spending and investment, influences overall economic activity and business performance. In the context of manufacturing firms, fluctuations in inflation and exchange rates directly impact the cost of inputs, production, and profitability, affecting aggregate demand and, consequently, financial performance. According to Keynesian principles, when inflation is high, purchasing power decreases, leading to reduced consumer demand, which can harm manufacturing companies by decreasing revenue and profitability (Keynes, 1936).

The theory's emphasis on government intervention is particularly relevant for economies like Nigeria, where structural and market vulnerabilities often exacerbate the effects of inflation and exchange rate volatility. Keynesian theory suggests that government intervention, such as adjusting interest rates or implementing fiscal policies, can stabilize prices and support manufacturing firms during economic downturns (Samuelson, 1948). Moreover, in environments where exchange rates fluctuate significantly, the uncertainty created may hinder investment decisions within the manufacturing sector. Keynesian insights suggest that stabilizing policies could reduce these uncertainties, enabling firms to better manage their operational costs and maintain financial stability (Hicks, 1937).

Furthermore, New Keynesian perspectives, which incorporate price stickiness and imperfect competition, highlight that manufacturing firms may not always adjust prices immediately in response to inflation or currency fluctuations due to long-term contracts or competitive pressures. This rigidity underscores the need for policy intervention to moderate inflationary pressures, allowing firms time to adjust without substantial losses (Mankiw, 1990). Thus, Keynesian Theory supports this study's focus on inflation and exchange rate impacts, as it provides an essential macroeconomic foundation for evaluating how these variables influence corporate financial performance and suggests potential avenues for policy responses to mitigate adverse effects on the manufacturing sector in Nigeria.

Research Methodology

The independent variables of this study are inflation rate and exchanges rate, while dependent variable is financial performance measured by Tobin's Q. The study adopted an Ex post facto research design. The justification for adopting ex post facto

research design is based on the fact that it is appropriate when the researcher does not have control over the independent variables, as they have already occurred and cannot be manipulated (Kerlinger & Lee, 2000). Also, an ex post facto design allows the researcher to collect and analyze this existing data to establish the relationships between the variables of interest (Kerlinger & Lee, 2000). The study covered selected forty-three (43) quoted manufacturing firms in Nigeria. the justification for using quoted manufacturing in this study is based on fact that they are publicly listed companies are required to disclose extensive financial and operational data, which is readily available from various databases and financial reporting sources. The study used panel data for the period of ten (10) years spanning from 2014 to 2023. The secondary data used for the study was extracted from annual financial reports, such as Statement of Comprehensive Income and the Statement of Financial Position, of the selected companies for the period 2014 to 2023 and from published annual reports of Statistical Bulletin of the Central Bank of Nigeria (CBN), National Bureau of Statistics (NBS) and Nigerian Stock Exchange (NSE) for the period of 2014 to 2023. This study used Regression analysis to test the hypotheses raised. The analysis was done with the aid of EVIEWS software.

The Regression Model Used

$$TBQ = \beta_0 + \beta_1 IFR + \beta_2 EXR + \mu$$

Where:

TBQ = Tobin Q

β_0 = Constant term, which represents when all explanatory variables are held constant

β_1 - β_2 = Coefficient of the parameter estimates

IFR = Inflation Rate

EXR = Exchange Rate

μ = Error term

Measurement and description of the Variables

Table 1. Variable Measurement

Variable Name & acronym	Variable type	Variable Description/Measurement	Sources
Tobin's Q (TBQ)	Dependent variable	Tobin's Q = (Market Value of Equity + Market Value of Debt) / Replacement Cost of Assets	Tobin (1969).
Inflation Rate	Independent variable	Inflation Rate = [(Current Price Level - Previous Price Level) / Previous Price Level] × 100	Kurihara. (2006).
Exchange Rate	Independent variable	Exchange Rate = (Local Currency Units / Foreign Currency Units)	Kurihara. (2006).

Source: Researcher's Compilation, 2024

Results and discussion

This section presents the results of the data analysis, followed by an in-depth discussion of the findings. Key outcomes are highlighted, with a thorough interpretation provided in alignment with the research objectives and hypotheses. Results are presented in a clear and structured manner, using tables and figures to enhance understanding. The discussion contextualizes the findings by comparing them with previous research and relevant theoretical perspectives from the literature. Furthermore, this section outlines the implications of the results for the field, addressing whether the hypotheses were supported or refuted. Practical and theoretical insights into the significance of the research outcomes are also discussed.

Table 2: Descriptive Statistics

	TBQ	IFR	EXR
Mean	0.315	0.874	0.690
Median	0.110	0.102	0.213
Maximum	0.376	4.221	4.144
Minimum	0.125	1.620	1.393
Std. Dev.	0.043	0.111	0.301
Skewness	0.193	0.150	0.149
Kurtosis	2.761	2.958	2.962
Jarque-Bera	3.668	0.210	0.188
Probability	0.224	0.877	0.755
Sum	31.50134	87,412.077	1,220.100
Sum Sq. Dev.	0.183000	69.001	8.970.677
Observations	430	430	430

Source: Researcher's Computation, 2024 via EVIEWS

The table 2 is the descriptive statistics for the variables used in this study. The table offer insights into the descriptive characteristics of the three variables: TBQ (Tobin's Q), IFR (Inflation Rate) and EXR (Exchange Rate) and covered the period of 10years from 2014 to 2023. The result shows the behaviour of the data.

The mean value of Tobin's Q (TBQ) is 0.315, with a relatively low standard deviation of 0.043, indicating limited variation in TBQ values across observations. The slight positive skewness (0.193) and a kurtosis close to 3 (2.761) suggest that the data distribution is approximately normal. A Jarque-Bera statistic of 3.668 with a probability of 0.224 further supports normality, as the probability exceeds the typical significance threshold (0.05). The sum of 31.50 and Sum Sq. Dev. of 0.183 reveal modest overall values and variability in TBQ for the sample. The implication is that TBQ values remain relatively stable, suggesting that firm market values, in relation to

their asset replacement costs, are not highly dispersed across observations, possibly indicating a consistent financial performance pattern in firms regarding their market valuation.

For the inflation rate, the average inflation rate (IFR) stands at 0.874, with a standard deviation of 0.111, indicating mild variability in inflation rates across observations. The median is 0.102, while the maximum is notably high at 4.221, suggesting occasional spikes in inflation rates. The skewness (0.150) and kurtosis (2.958) values approximate normality, supported by a Jarque-Bera probability of 0.877, which exceeds 0.05. The sum of IFR values is 87,412.077, and the Sum Sq. Dev. is 69.001, reflecting considerable cumulative inflation impact over the sample period. This indicates that inflation rates vary moderately but generally align with a normal distribution, which could suggest relatively stable but occasionally volatile inflationary conditions affecting firm costs and overall market prices.

For the exchange rate, the mean exchange rate (EXR) is 0.690, with a standard deviation of 0.301, indicating higher variability relative to TBQ and IFR. The median value of 0.213, a maximum of 4.144, and a minimum of 1.393 suggest a wide range, with occasional extreme values. The skewness (0.149) and kurtosis (2.962) are near-normal, and the Jarque-Bera probability of 0.755 implies a normal distribution, as it exceeds 0.05. With a sum of 1,220.100 and Sum Sq. Dev. of 8.970.677, the data indicates substantial variability and overall magnitude in exchange rates. The implication is that exchange rate fluctuations are significant and potentially impactful on manufacturing firms' financial performance, especially due to the high variability, which may influence costs for imported materials, ultimately affecting profitability and stability.

Table 3: Correlation of the Variables

Date: 10/11/24 Time: 07:55

Sample: 2014 2023

Included observations: 430

Number of group: 43

Correlation Probability	TBQ	IFR	EXR
TBQ	1.000		
IFR	-0.124 0.000	1.000	
EXR	0.196 0.000	0.421 0.000	1.000

r=correlation coefficient; t -stat; probability of t -statistics .

Source: Researcher's Computation, 2024 via EVIEWS

The table 3 is a correlation analysis. The correlation matrix provided indicates the relationships between Tobin's Q (TBQ), IFR (Inflation Rate) and EXR (Exchange Rate) based on a sample period from 2014 to 2023, with 430 observations from 43 manufacturing firms. The correlation coefficients measure the strength and direction of the linear relationships between the variables, while the probabilities (p-values) help determine the statistical significance of these relationships.

The correlation coefficient between TBQ (Tobin's Q) and IFR (Inflation Rate) is -0.124, with a probability value of 0.000. This indicates a weak negative relationship that is statistically significant. Although the relationship is statistically significant (as shown by the probability of 0.000), the strength is minimal, indicating that inflation rate changes have a weak inverse impact on Tobin's Q for the firms in the sample.

The correlation between TBQ and EXR (Exchange Rate) is 0.196, with a probability of 0.000. This positive but weak correlation is statistically significant, suggesting a slight tendency for Tobin's Q to increase as the exchange rate rises. While significant, the strength of this association is low, implying that exchange rate fluctuations have a limited positive impact on Tobin's Q. This might reflect that firm valuations are somewhat sensitive to currency value, possibly due to import or export cost effects, but the effect is not strong.

The correlation between IFR and EXR is 0.421, with a probability of 0.000, indicating a moderate positive relationship that is statistically significant. This moderate correlation suggests that as inflation increases, exchange rates tend to increase as well, possibly due to inflationary pressures leading to currency devaluation. This relationship implies that inflation rate fluctuations are more strongly associated with changes in the exchange rate than with Tobin's Q, highlighting a moderate co-movement between inflation and currency value.

Regression Result

The table 4. present the regression result for the effect of inflation and exchange rates on financial performance (proxy by Tobin's Q) of the quoted selected manufacturing companies in Nigeria.

Model Fit (R-squared and Adjusted R-squared): The R-squared value is 0.278248, and the Adjusted R-squared is 0.241638. These values indicate that approximately 24.2% of the variability in financial performance proxy by Tobin's Q (TBQ) is explained by inflation rate (IFR) and exchange rate (EXR) while around 75.8% of the variation remains unexplained by this model. While this is a modest fit, it suggests that other factors not included in the model may also significantly impact Tobin's Q. This moderate explanatory power implies the need for further research to capture additional determinants of firm valuation. Although the model is statistically significant (as indicated by the F-statistic and its p-value), the moderate R-squared and Adjusted R-squared values suggest that the model does not fully capture all determinants of Tobin's Q. In practical terms, this does not invalidate the model; it means the model is limited in its explanatory power, capturing only part of the underlying dynamics affecting financial performance.

Statistical Significance of the Model (F-statistic and Prob(F-statistic)): The F-statistic is 7.600210, with a probability of 0.00000, indicating that the overall regression model is statistically significant. This significance shows that, as a whole, IFR and EXR jointly influence Tobin's Q at a meaningful level, supporting the model's utility in explaining some degree of variance in firm valuation.

The S.E. of regression value of 0.213122 measures the typical distance between the actual and predicted values of TBQ. A smaller value indicates that the model fits the data reasonably well. However, given the low R-squared value, the model still leaves room for improvement in terms of explaining variation in TBQ.

The Durbin-Watson statistic is a measure used to detect autocorrelation, or serial correlation, in the residuals of a regression model. Autocorrelation occurs when the residuals (errors) from one observation are correlated with residuals from another, which violates a key assumption of linear regression: the independence of errors. When autocorrelation is present, it can indicate that important variables or patterns are not accounted for in the model, potentially biasing coefficient estimates and making them less reliable. In this model, the Durbin-Watson statistic is 1.762331, which is close to 2. A Durbin-Watson value near 2 generally suggests that there is minimal autocorrelation in the residuals. Values significantly below 2 indicate positive autocorrelation, while values above 2 indicate negative autocorrelation. Since the result here is close to 2, we can conclude that the model's residuals do not exhibit serious autocorrelation. The absence of significant autocorrelation implies that the model's estimates are likely to be more reliable and efficient, as the independence of errors assumption is largely met. This strengthens the validity of the regression coefficients, suggesting they accurately reflect the relationships between the dependent variable (Tobin's Q) and the independent variables (IFR and EXR).

Table 4: Regression Result

Dependent Variable: TBQ

Method: Least Squares

Date: 10/11/24 Time: 08:12

Sample: 2014 2023

Included observations: 430

Group: 43

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	25.80579	773.9502	0.033343	0.9737
IFR	-0.716011	0.224552	-2.712122	0.0138
EXR	-4993.444	11.35.591	- 3.519396	0.0006
R-squared	0.278248	Mean dependent var		0.12906
Adjusted R-squared	0.241638	S.D. dependent var		0. 63938
S.E. of regression	0.423122	Akaike info criterion		1. 27924
Sum squared resid	24.71646	Schwarz criterion		1. 66319
Log likelihood	-238.7697	Hannan-Quinn criter.		1. 39341
F-statistic	7.600210	Durbin-Watson stat		1.762331
Prob(F-statistic)	0.00000			

Source: Researcher's Computation, 2024 via EVIEWS

From the table 4, The constant term (C) has a coefficient of 0.259721 with a p-value of 0.2063. This suggests that when both GDP and Interest Rate (INT) are zero, Tobin's Q (TBQ) is expected to be around 0.259. However, the p-value of 0.2063 is greater than 0.05, indicating that the constant term is not statistically significant. This means that the intercept does not provide strong evidence that TBQ is significantly different from zero when GDP and INT are zero, which is a common scenario in many macroeconomic models where the constant term is not critical.

Test of Hypotheses and Discussion of Result

H₀₁: Inflation rate has no significant effect on financial performance (Tobin's Q) of selected quoted manufacturing companies in Nigeria

The coefficient for inflation rate (IFR) is -0.716011, with a t-statistic of -2.712122 and a probability value of 0.0138, making it statistically significant at conventional (5%) levels. This result revealed that inflation rate has negative and significant effect on financial performance (Tobin's Q) of selected quoted manufacturing companies in Nigeria. The negative coefficient suggests that as the inflation rate increases, Tobin's Q tends to decrease, reflecting an inverse relationship between inflation and firm valuation. This result implies that higher inflation can erode the market valuation of firms, possibly by increasing production costs and reducing profit margins. The significance of this variable supports the importance of controlling inflation to enhance firm value.

The practical implication of this result is that rising inflation in Nigeria can significantly harm the financial performance and market valuation of manufacturing companies. The negative coefficient for the inflation rate (IFR) suggests that higher inflation correlates with lower Tobin's Q values, indicating that as inflation increases, the relative market valuation of firms tends to decline. This inverse relationship implies that inflation pressures erode firm value by increasing production costs and squeezing profit margins. For manufacturing companies, this result highlights the financial strain caused by inflation, especially when costs for materials, labor, energy, and other inputs rise faster than product prices can be adjusted. High inflation makes it more challenging for firms to maintain profitability, as passing costs onto consumers may reduce demand for their products. Additionally, inflation can raise borrowing costs, increase uncertainty, and lead to volatility in business planning, making it more difficult for firms to forecast expenses and revenue.

This finding is in line with the finding of Alimi (2014) investigated inflation's long-term impact on Nigeria's financial sector, covering data from 1970 to 2012 and found that inflation negatively affects financial sector performance.

H₀₂: Exchange rate has no significant effect on financial performance (Tobin's Q) of manufacturing companies in Nigeria

The coefficient for EXR is -4993.444, with a t-statistic of -3.519396 and a probability of 0.0006, indicating statistical significance at 5% level. This result revealed that exchange rate has negative and significant effect on financial performance (Tobin's Q) of selected quoted manufacturing companies in Nigeria. The large negative coefficient implies a strong inverse relationship between exchange rate changes and Tobin's Q, suggesting that as the exchange rate (likely indicating currency depreciation) increases, firm valuation declines substantially. This result highlights the sensitivity of manufacturing firms to exchange rate fluctuations, as a weaker currency could increase input costs for firms reliant on imported goods, reducing profitability and market valuation. This finding underscores the importance of exchange rate stability for firm performance.

The practical implication of this result is that exchange rate fluctuations, particularly depreciation of the local currency, can significantly undermine the financial performance and market valuation of manufacturing firms in Nigeria. With the large negative coefficient for EXR, the study indicates that as the exchange rate rises (implying currency depreciation), Tobin's Q – a measure of firm value relative to asset replacement cost – declines sharply. This relationship suggests that manufacturing firms in Nigeria are highly sensitive to exchange rate instability. Practically, this finding means that when the Nigerian Naira depreciates against other currencies, manufacturing firms face higher costs for imported raw materials, machinery, and other production inputs. Since these firms rely on foreign inputs, a weaker currency can sharply increase their production costs, erode profit margins, and reduce their overall valuation in the market. Firms may be forced to either absorb these higher costs, which would lower profitability, or pass them onto consumers, potentially reducing demand for their products.

This finding is in line with the finding of Olabode (2018) analyzed the effect of foreign exchange rates on the performance of Nigeria's manufacturing sector from 1990 to 2016 and found significant negative impacts of exchange rates, inflation, and interest rates on manufacturing performance. The finding also agreed with the finding of Majumdar (2017) explored the effects of macroeconomic variables, particularly exchange rates, on the financial performance of Nigerian firms and found that exchange rates adversely affected firms by increasing operational costs, especially for companies reliant on imports.

This study found that both inflation and exchange rates have negative and significant effect on financial performance (Tobin's Q) of the selected quoted manufacturing firms in Nigeria at 5% level of significant.

Conclusions and Recommendation

Based on the findings, this study concludes that both inflation and exchange rates positively and significant influences the financial performance as measured by Tobin's Q of quoted manufacturing firms in Nigeria, the negative coefficients for both

variables indicate that as inflation and exchange rates rise, firm valuation tends to decline. This inverse relationship suggests that inflation increases production costs, erodes profit margins, and reduces firm value, while exchange rate depreciation heightens input costs, particularly for firms reliant on imported materials, further lowering market valuation. These results underscore the importance of macroeconomic stability particularly in controlling inflation and maintaining exchange rate stability to enhance the financial health and competitiveness of Nigeria's manufacturing sector. This study therefore recommends based on the findings and conclusion that:

- i. Manufacturing firms in Nigeria should strengthen their resilience to rising inflation-related production costs by enhancing operational efficiency through energy-efficient technologies and waste reduction. They can also secure long-term contracts with suppliers to stabilize input prices and explore local sourcing to decrease reliance on imports affected by inflation. These strategies can help protect profit margins from inflation's adverse effects.
- ii. Manufacturing firms in Nigeria should implement strong foreign exchange risk management strategies, such as hedging with forward contracts or options, to protect against exchange rate volatility. Gradually diversifying supply chains by sourcing materials locally can also reduce reliance on imports affected by currency fluctuations. These measures help firms manage costs, preserve profit margins, and maintain financial stability, even amid currency depreciation.

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